2026 IEEE 5th International Conference on Robotics, Automation,
Artificial-Intelligence and Internet-of-Things (RAAICON)

*18–19 September 2026 · Jashore University of Science and Technology, Jashore, Bangladesh*

# RGEF: Reliability-Gated Evidential Fusion for Robust Medical AI

*Out-of-distribution detection for dermatological artificial intelligence*

**Md Abu Sayeb Rayhan**[1] Mosammat Fahimah[1] Md. Tarek Mahmud[1] Hustagna Hiya[2] Ajoy Shutra Dhar[1]

[1] Department of Computer Science & Engineering, Comilla University, Kotbari, Cumilla-3506, Bangladesh

[2] Institute of Information Technology, Jahangirnagar University, Savar, Dhaka-1342, Bangladesh

Paper ID 487 · Oral presentation

979-8-3195-3824-6/26/$31.00 ©2026 IEEE

OUTLINE

# Outline of the talk

| | | |
|---|---|---|
| 1 | **Motivation** | *Silent failure under distribution shift; why a single uncertainty estimator is not enough* |
| 2 | **Objectives and contributions** | *What the method is required to do, and the three claims we defend* |
| 3 | **Method** | *Reliability-gated evidential fusion, foundation-model routing, off-support fallback* |
| 4 | **Experimental setup** | *One internal and two strictly zero-shot external benchmarks* |
| 5 | **Results and analysis** | *Comparison with eleven published detectors; transfer, mechanism, error audit* |
| 6 | **Limitations** | *Where the method does not work, and what localises the failure* |
| 7 | **Conclusions** | *Two lessons that carry beyond dermatology* |

> **Central claim.** *Reliability-aware fusion — rather than any individual estimator — is the key ingredient for safe dermatology AI.*

*Every figure and every number shown is reproduced from the camera-ready manuscript.*

§1 MOTIVATION

# Silent failure under distribution shift

Deep skin-lesion classifiers rival dermatologists on curated benchmarks. That accuracy presumes test images are drawn from the training distribution — a premise clinical practice violates daily. Two failure modes are of particular concern.

### Covariate shift

P(x) changes while the label space does not: skin phototype, dermoscope versus smartphone camera, illumination, hair, rulers, preprocessing.

Calibration deteriorates well before accuracy does.

### Semantic shift

Classes never seen in training. HAM10000 contains no squamous cell carcinoma, although SCC is common in practice and supplies 628 images in ISIC 2019.

Lacking any mechanism to signal novelty, a closed-world classifier assigns such an input to the nearest familiar class.

In either case the network still returns a confident prediction — routinely with softmax above **0.99**. What follows is not a benign error but a missed malignancy or an unnecessary excision, and no flag is ever raised.

*A safe system must be able to say "this input is unfamiliar" — for both kinds of shift, and before it commits to a diagnosis.*

§1 MOTIVATION

# Every single uncertainty estimator has a blind spot

Each paradigm is well founded; each fails in a documented, different way.

| Paradigm | Uncertainty signal | Documented failure mode |
|---|---|---|
| **Deep ensembles** | Mutual information between predictions and parameters — member disagreement. | Under dataset shift the members share the same corrupted features and agree confidently on the wrong answer, so disagreement collapses even as error rises. |
| **Evidential deep learning** | Subjective-logic vacuity from a single-pass Dirichlet. | Vacuity is a function of those same features. When the backbone shifts, evidence is assigned confidently to the wrong class and vacuity reads low exactly when it should read high. |
| **Conformal prediction** | RAPS prediction-set size, with a finite-sample coverage guarantee. | The guarantee is marginal and rests on exchangeability — both of which shift violates — so a far-OOD input can still fall in a small, confident set. |

> **Crucially, these blind spots do not overlap.** Robustness therefore calls not for a stronger estimator, but for a principled way of combining estimators that recognises when each can be trusted.

**Research question.** *A single fusion weight cannot do this: reliability varies from input to input. Can it be learned per sample?*

# Objectives and contributions

> **Objective.** Detect both semantic and covariate shift in dermatology with a single calibrated score — one reliable enough to drive a clinical decision, and provably never worse than the static fusion it generalises.

## C1 Reliability-gated fusion

We cast OOD detection as input-conditional, reliability-discounted fusion of P = 8 complementary pillars — ensemble mutual information, evidential vacuity, conformal set size and five feature-space detectors — mediated by a learned gate. Static and scalar fusion are recovered exactly as the constant-gate special case, which gating dominates whenever the strongest pillar changes across inputs.

## C2 Foundation-model routing

The gate is conditioned on frozen DINOv2 embeddings rather than on the task backbone. Cross-entropy representations collapse toward class prototypes under shift; self-supervised geometry retains a smoother, domain-invariant structure. DINOv2 features are weak detectors in isolation but dependable routers — and conditioning on them improves every dataset we test.

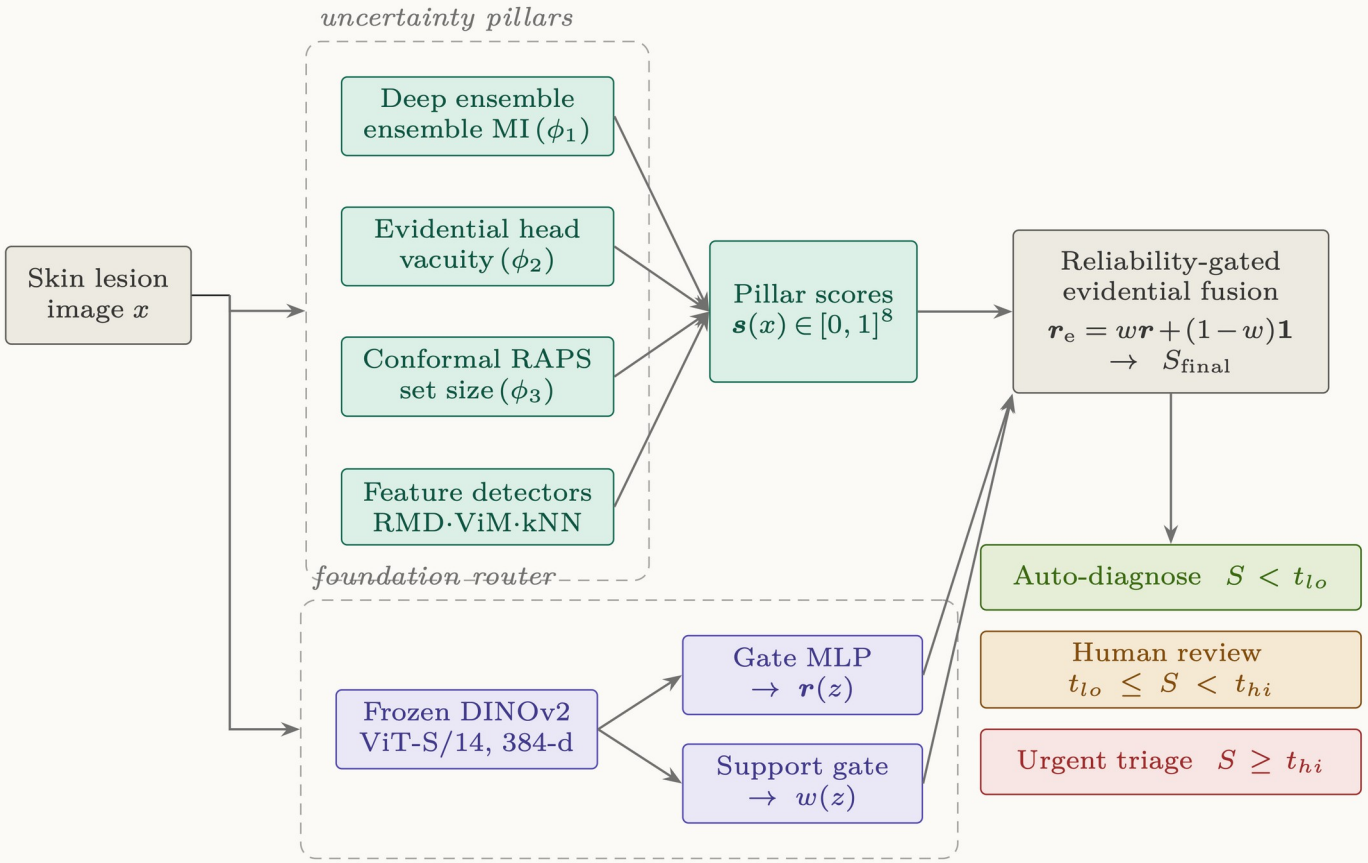
## C3 Off-support safety guard

A density-based fallback identifies inputs beyond the gate's calibrated support and smoothly reverts them to static fusion, so adaptivity is bounded below rather than unbounded — a floor RGEF cannot fall through even under severe shift.

*Validation: eleven published post-hoc detectors, one internal near-OOD split and two strictly zero-shot external datasets covering both shift types, ten seeds, DeLong tests, a per-class error audit and a three-tier triage protocol.*

§3 METHOD

# Architecture




## Eight pillars

Three uncertainty estimators — ensemble MI, evidential vacuity and conformal set size — plus five feature-space detectors: RMD, ViM, deep k-NN, DINOv2-RMD and DINOv2-kNN.

## Foundation router

Frozen DINOv2 ViT-S/14, 384-d, reduced to 64-d by PCA. Used only for routing and support estimation — never as a detector, and never fine-tuned.

## Calibrated output

One fused score per image, mapped through three clinical tiers: auto-diagnose, human review, urgent triage.

**Fig. 1.** *RGEF. Eight uncertainty pillars are fused under per-sample reliabilities produced by a gate routed on frozen DINOv2 geometry and guarded by a density-based support weight; the calibrated score drives a three-tier triage decision.*

# Reliability-gated evidential fusion

## Fusion

Each pillar score is min–max normalised to $s_j \in [0,1]$ and recast as a binary subjective opinion with base vacuity $u_0$:

$$b^j_O = s_j\,(1-u_0), \quad b^j_I = (1-s_j)(1-u_0) \tag{1}$$

A two-layer MLP maps the frozen DINOv2 embedding z to one reliability per pillar. The sigmoid acts per pillar, so the $r_j$ are Shafer discount factors, not a simplex:

$$\boldsymbol{r}(z) = \sigma(\, g_\psi(z)\,) \in (0,1)^P \tag{2}$$

Reliability-discounted evidence accumulates into a Beta posterior whose mean is the integrated OOD score:

$$\mathcal{E}_O = \Sigma_j\, r_j\, b^j_O, \quad \mathcal{E}_I = \Sigma_j\, r_j\, b^j_I \tag{3}$$

$$S_{OOD} = (\mathcal{E}_O + 1) \,/\, (\mathcal{E}_O + \mathcal{E}_I + 2) \tag{4}$$

> **Setting $r_j \equiv 1$ recovers gate-free static fusion.** The gate is exactly the input-conditional novelty — nothing else in the pipeline changes.

## Off-support guard

An in-support weight is derived from the distance to the k-th nearest training neighbour (k = 50) in $\ell_2$-normalised DINOv2 space, with $\rho_{hi}$ the 99th percentile of training distances:

$$w(z) = \exp(\, -[\, d_k(z) - \rho_{hi}\,]^+ / \tau\,) \tag{5}$$

$$\boldsymbol{r}_{eff}(z) = w(z)\,\boldsymbol{r}(z) + (1 - w(z))\,\mathbf{1} \tag{6}$$

As an input drifts from the calibration manifold, w declines and the model degrades gracefully to static fusion rather than extrapolating.

## Training objective

Class-balanced binary cross-entropy, an $\ell_1$ sparsity penalty on the reliabilities, and a hinge that charges the fused score whenever it trails the strongest individual pillar:

$$\mathcal{L}(\psi) = \mathrm{BCE}_w(S, y) + \lambda\|\boldsymbol{r}\|_1 + \beta[\, \mathrm{BCE}_w(S, y) - \min_j \mathrm{BCE}_w(s_j, y)\,]^+ \tag{7}$$

Fitted by five-fold cross-validation. The hinge and the guard together bound the worst case from below.

§4 SETUP

# Benchmarks: one internal, two strictly zero-shot

| Class | Role | HAM10000 | ISIC 2019 | PAD-UFES |
|---|---|---|---|---|
| nv | ID | 6,705 | 12,875 | — |
| mel | ID | 1,113 | 4,522 | — |
| bkl | ID | 1,099 | 2,624 | — |
| bcc | ID | 514 | 3,323 | — |
| akiec | ID | 327 | 867 | — |
| df | OOD | 115 | 239 | — |
| vasc | OOD | 142 | 253 | — |
| scc | OOD | — | 628 | 192 |
| **ID total** | | **9,758** | **24,211** | **2,106** |
| **OOD total** | | **257** | **1,120** | **192** |

**Table 1.** *Class-wise composition after taxonomy alignment. HAM10000 ID images are split 70/15/15 class-stratified (6,830 / 1,464 / 1,464, seed 42). PAD-UFES-20 ID images map onto the five-class space only in aggregate.*

> **df and vasc are removed before splitting**, so no OOD image ever enters classifier training. No target-domain image enters training or calibration — both external evaluations are strictly zero-shot, with every model frozen.

## Protocol

### Ensemble

M = 5 ImageNet-pretrained backbones — ResNet-50, EfficientNet-B0, DenseNet-121, ResNet-34, MobileNet-V2 — at 224×224.

### Router

Frozen DINOv2 ViT-S/14 (384-d); the supervised-gate ablation substitutes 2048-d ResNet-50 features.

### Conformal

Split RAPS at $\alpha = 0.10$: mean set size 2.33 at 99.4% empirical coverage.

### ID performance

93.31% accuracy, macro-F1 0.893; ECE 0.0369, tightening to 0.0140 under temperature scaling.

### Baselines

Eleven published post-hoc detectors across three families, the three individual pillars, and four fusion baselines.

### Statistics

AUROC, AUPR and FPR@95; DeLong tests on correlated ROC curves; ten seeds with 95% confidence intervals.

# Internal near-OOD benchmark: eleven published detectors

| Method | AUROC ↑ | AUPR ↑ | FPR@95 ↓ |
|---|---|---|---|
| *Logit- and activation-based* | | | |
| MSP | 0.7805 | 0.3706 | 78.0 |
| Energy | 0.7792 | 0.3986 | 80.9 |
| ASH-S | 0.4058 | 0.1389 | 96.2 |
| *Feature space* | | | |
| Deep k-NN | 0.7550 | 0.3194 | 63.6 |
| DINOv2-RMD | 0.5542 | 0.2523 | 96.2 |
| *Single uncertainty pillars* | | | |
| Evidential vacuity (best pillar) | 0.8179 | 0.4105 | 62.4 |
| *Fusion baselines* | | | |
| Static evidential ($r \equiv 1$) | 0.8195 | 0.4382 | 52.6 |
| Scalar meta-learner | 0.8239 | 0.4405 | 53.6 |
| *Ours* | | | |
| RGEF (ResNet routing) | 0.9136 | 0.6447 | 37.4 |
| **RGEF (DINOv2 routing, with fallback)** | **0.9560** | **0.7923** | **23.2** |

**Table 2.** *HAM10000 near-OOD benchmark, df + vasc held out (1,464 ID / 257 OOD). Representative rows of the eleven baselines; FPR@95 in %. Single-run pillar-deconstruction protocol — the gate-ablation protocol of Table 3 scores 0.9594, and the 10-seed mean is 0.9596 ± 0.0018.*

## Three patterns

1. Logit- and activation-based scores transfer poorly to fine-grained dermatology. ASH-S falls below chance (0.4058): pruning late-layer activations discards precisely the texture that separates df and vasc from nv.

2. Fixed-weight and scalar combiners (0.8118–0.8239) barely exceed the best single pillar (0.8179) — fixed weights cannot exploit complementarity.

3. Input-dependent gating breaks the pattern, and routing on frozen DINOv2 improves it further: best AUROC and AUPR at the lowest FPR@95.

**52.6 % → 23.2 %**

FPR@95 against static fusion. DeLong tests confirm significance over both the strongest pillar and static fusion, $p < 10^{-4}$.

§5 RESULTS

# Zero-shot transfer under external shift

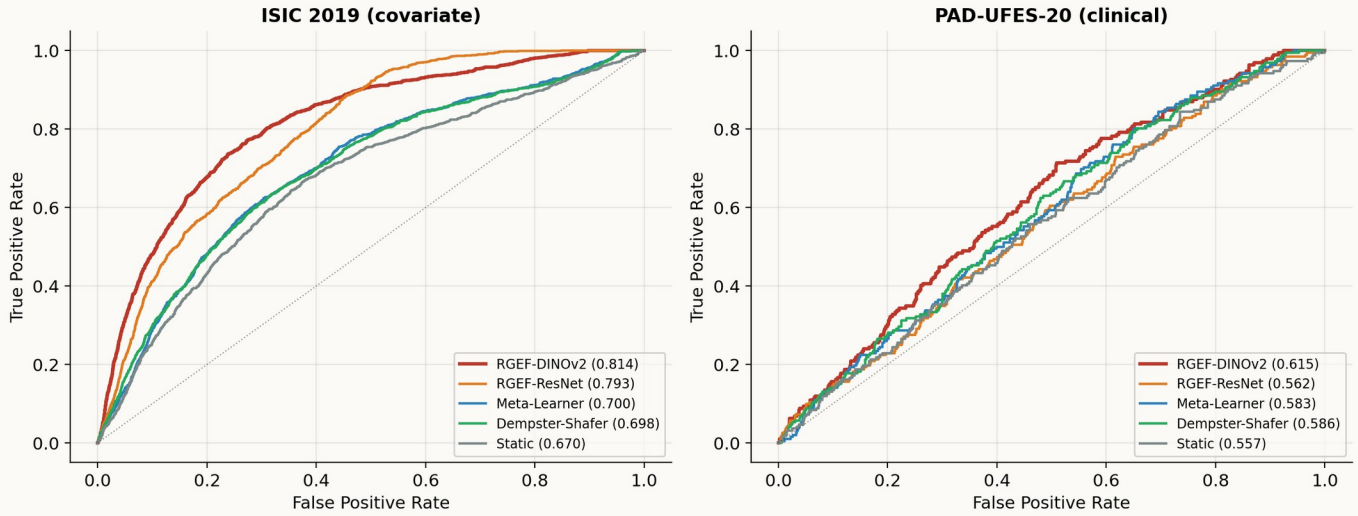



**Fig. 2.** *ROC under external shift, all models frozen after HAM10000 training. DINOv2 routing sustains a higher true-positive rate at low false-positive rate.*

## Robustness validation

**Seed variance** — 0.9596 ± 0.0018 over ten gate retrainings (95% CI [0.9585, 0.9608]); 0.9125 ± 0.0014 for the ResNet route — roughly an order of magnitude tighter than the baselines' bootstrap intervals.

**Paired significance** — DeLong tests on correlated ROC curves, reported throughout.

**Distributional** — Two external datasets evaluated zero-shot, spanning semantic and modality shift.

| Fusion strategy | HAM | ISIC | PAD |
|---|---|---|---|
| Static ($r \equiv 1$) | 0.8195 | 0.6705 | 0.5567 |
| Scalar meta-learner | 0.8239 | 0.6996 | 0.5828 |
| RGEF (ResNet gate) | 0.9174 | 0.7926 | 0.5621 |
| **RGEF (DINOv2 gate)** | **0.9594** | **0.8144** | **0.6148** |

**Table 3.** *Gating-feature ablation, AUROC. Performance rises monotonically from static fusion to ResNet routing to DINOv2 routing on all three benchmarks.*

### ISIC 2019

Static fusion collapses to 0.6705 at 90.3% FPR@95 — at high recall almost every outlier, SCC included, slips through. The full DINOv2 gate reaches 0.8144: +0.144 over static fusion and +0.022 over the ResNet gate, both $p < 10^{-4}$.

### The guard rarely fires

Mean support weight stays at 1.00 / 1.00, 0.98 / 0.96 and 0.96 / 0.95 for ID / OOD inputs. The gain is attributable to the gate; the guard acts as a worst-case floor, not as the mechanism.

§5 MECHANISM

# Foundation embeddings are routers, not detectors

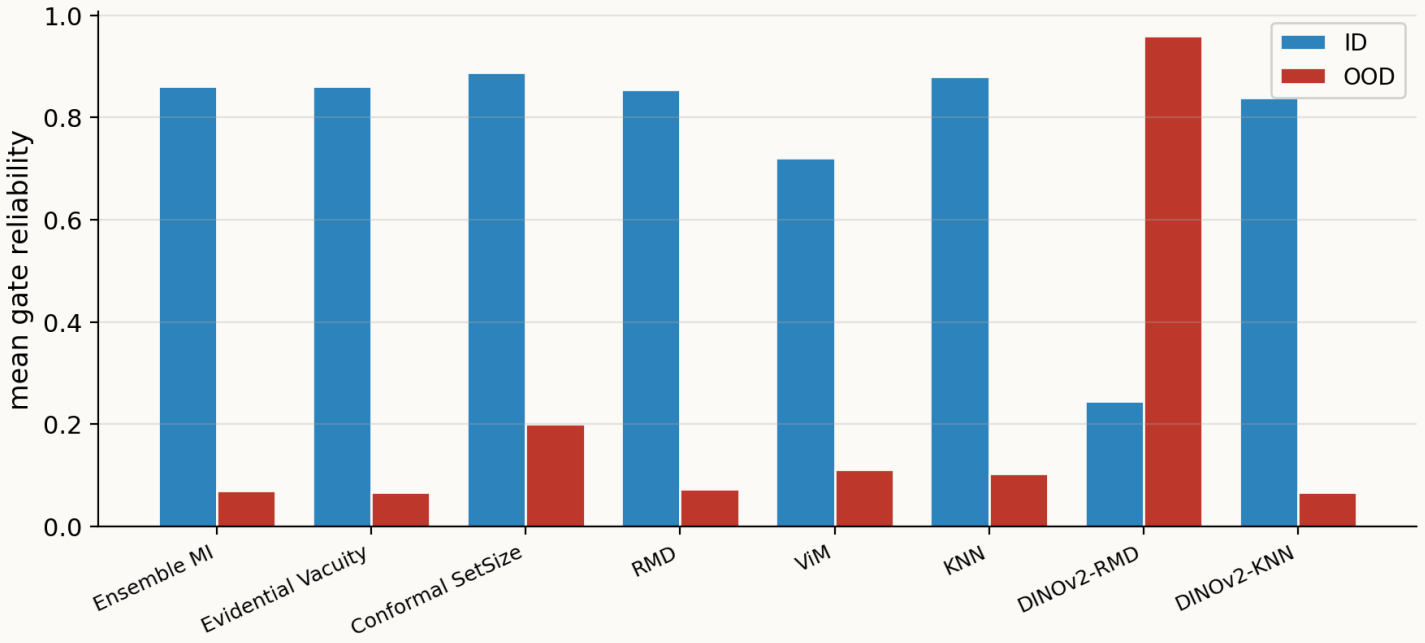



**Fig. 3.** *Learned per-pillar reliabilities on ID and OOD inputs. The gate suppresses the task-backbone pillars on OOD data and shifts trust onto the DINOv2-space detector, whose own AUROC is near chance.*

*A router need only place an input faithfully; it does not need to score it. This dissociation is not a rule we imposed — it is what the gate learns, and it is what motivates the reliability-gated design.*

## The same embedding, two roles

| DINOv2-RMD | HAM | ISIC | PAD |
|---|---|---|---|
| as detector | 0.5542 | 0.3856 | 0.4227 |
| **as router** | **0.9594** | **0.8144** | **0.6148** |

*AUROC. Two of the three standalone scores fall below chance; routing on the very same features gives the best detector on every benchmark.*

**0.285 → 0.932**

The gate inverts its trust on OOD inputs: it raises the DINOv2-RMD reliability while suppressing every task-backbone pillar.

*Used directly as detectors, DINOv2 scores are weak; used to estimate reliability, the same geometry is decisive.*

§5 ANALYSIS

# Error analysis and clinical triage

| Class | Role | n | Mean score | Flag % | Urgent % |
|---|---|---|---|---|---|
| **scc (novel)** | **OOD** | **628** | **0.231** | **26.3** | **22.3** |
| df | OOD | 239 | 0.223 | 23.0 | 19.7 |
| vasc | OOD | 253 | 0.195 | 18.2 | 14.2 |
| all ID classes | ID | 24,211 | — | 8.1 | 7.2 |

**Table 4.** *Per-class outcomes on ISIC 2019 under the triage rule opposite. "Flagged" = review or urgent.*

### SCC — the genuinely novel class — is best detected

A semantic novelty with no nearby in-distribution prototype drives all pillars up together, so the fused score rises and the case escalates — precisely the behaviour required of a safety layer.

### vasc is the residual failure mode

Vascular lesions are chromatically distinctive but morphologically simple, so the classifier maps them confidently onto nv / mel and all three core pillars agree — the jointly-overconfident regime no fusion of those pillars alone can resolve, and the reason the DINOv2-space detectors are retained as extra pillars.

## Three-tier triage protocol

**I** **Auto-diagnose** $S < t_{lo}$

Confident and in-distribution; handled by the model.

**II** **Human review** $t_{lo} \le S < t_{hi}$

Uncertain but on-support; escalated to a clinician.

**III** **Urgent triage** $S \ge t_{hi}$

Highly uncertain or off-support; expert required.

**Pre-specified, not tuned.** $t_{lo} = 0.35$ and $t_{hi} = 0.55$, plus a 2-of-3 pillar-agreement override so that no single miscalibrated pillar can trigger or suppress an escalation alone. Fixed on the internal benchmark, then applied unchanged to ISIC 2019.

*Caveat: the audit is computed on the triple-uncertainty score for which this rule is defined. Re-deriving the thresholds on the RGEF scale is future work — but a uniformly higher ROC dominates, so every achievable safety / workload trade-off improves.*

§6 LIMITATIONS

# Limitations and threats to validity

The principal weakness is the dermoscopy-to-smartphone gap of PAD-UFES-20, stated plainly rather than dropped.

## PAD-UFES-20: the modality gap

RGEF improves on static fusion (0.5567 → 0.6148, $p = 1.5\times10^{-4}$) and on the ResNet gate ($p = 2.1\times10^{-3}$), but its margin over the strongest scalar baselines is **not significant** — Dempster–Shafer 0.5856, $p = 0.099$; meta-learner 0.5828, $p = 0.077$.

**At an AUROC near 0.61 the detector is not clinically usable on smartphone imagery, and we make no claim otherwise.**

## The failure is localised, not diffuse

On this dataset DINOv2-RMD scores 0.4227 as a standalone detector — below chance. The router's own geometry has collapsed, so the gate can only arbitrate among uninformative signals. The bottleneck is the router's domain coverage, not the fusion mechanism — which points to a dermatology-specific foundation encoder, framework unchanged.

**Meanwhile the off-support floor still guarantees performance no worse than static fusion.**

### Further limitations

- Inference is dominated by the five-model ensemble (≈104M parameters, ≈21 GFLOPs per image), motivating distillation.
- ISIC 2019 overlaps HAM10000 in provenance, so its numbers are near- rather than fully-independent replication.
- The triage thresholds require site-specific re-calibration, and have not yet been re-derived on the RGEF scale.
- All evaluation is retrospective. Prospective, multi-centre validation across skin phototypes is required before deployment; no claim of clinical readiness is made.

# Conclusions

RGEF fuses ensemble divergence, evidential vacuity and conformal set size through an input-conditional gate routed on frozen foundation-model geometry and guarded by a density-based off-support fallback.

### 1 Input-conditional gating dominates static and scalar fusion

whenever pillar reliability varies with the input — and in dermatology it does. 0.9594 internal near-OOD AUROC at 23.2% FPR@95, $p < 10^{-4}$ over both the best pillar and static fusion.

### 2 Foundation embeddings are far more useful as routers than as detectors

for fine-grained novelty. This is the lesson that carries beyond dermatology: a weak anomaly score can still be an excellent routing substrate, because routing asks a different question of the geometry.

### 3 A worst-case floor is what makes adaptivity deployable

The off-support guard and the ≥ best-pillar hinge bound degradation, so the adaptive system provably cannot underperform the static one it generalises — zero-shot, 0.6705 → 0.8144 on ISIC 2019 and 0.5567 → 0.6148 on PAD-UFES-20.

**Future work**

A domain-matched router · thresholds re-derived on the RGEF scale · ensemble distillation · prospective multi-centre validation across skin phototypes

> ***Reliability-aware fusion, rather than any individual estimator, is the key ingredient for safe dermatology AI.***

REFERENCES

# Selected references


[1] A. Esteva et al. *Dermatologist-level classification of skin cancer with deep neural networks. Nature, 542:115–118, 2017.*

[2] B. Lakshminarayanan, A. Pritzel, C. Blundell. *Simple and scalable predictive uncertainty estimation using deep ensembles. NeurIPS, 2017.*

[3] Y. Ovadia et al. *Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. NeurIPS, 2019.*

[4] M. Sensoy, L. Kaplan, M. Kandemir. *Evidential deep learning to quantify classification uncertainty. NeurIPS, 2018.*

[5] A. N. Angelopoulos, S. Bates, M. Jordan, J. Malik. *Uncertainty sets for image classifiers using conformal prediction. ICLR, 2021.*

[6] M. Oquab et al. *DINOv2: learning robust visual features without supervision. TMLR, 2024.*

[7] D. Hendrycks, K. Gimpel. *A baseline for detecting misclassified and out-of-distribution examples in neural networks. ICLR, 2017.*

[8] W. Liu, X. Wang, J. Owens, Y. Li. *Energy-based out-of-distribution detection. NeurIPS, 2020.*

[9] Y. Sun, Y. Ming, X. Zhu, Y. Li. *Out-of-distribution detection with deep nearest neighbors. ICML, 2022.*

[10] H. Wang, Z. Li, L. Feng, W. Zhang. *ViM: out-of-distribution with virtual-logit matching. CVPR, 2022.*

[11] J. Ren et al. *A simple fix to Mahalanobis distance for improving near-OOD detection. arXiv:2106.09022, 2021.*

[12] A. Djurisic et al. *Extremely simple activation shaping for out-of-distribution detection. ICLR, 2023.*

[13] P. Tschandl, C. Rosendahl, H. Kittler. *The HAM10000 dataset. Scientific Data, 5:180161, 2018.*

[14] M. Combalia et al. *BCN20000: dermoscopic lesions in the wild. arXiv:1908.02288, 2019.*

[15] A. G. C. Pacheco et al. *PAD-UFES-20: a skin lesion dataset composed of patient data and clinical images. Data in Brief, 2020.*

[16] G. Shafer. *A Mathematical Theory of Evidence. Princeton University Press, 1976.*


*Sixteen of the thirty-one references cited in the camera-ready manuscript; the remainder appear there in full.*

# Thank you

*Questions and discussion are welcome.*

**RGEF: Reliability-Gated Evidential Fusion for Robust Medical AI**

Md Abu Sayeb Rayhan · Comilla University, Bangladesh
sayeb.cse@gmail.com

**Takeaway**

*Input-conditional gating dominates static fusion; foundation embeddings are routers, not detectors; a worst-case floor is what makes adaptivity safe to deploy.*

IEEE RAAICON 2026 · Paper ID 487 · Jashore University of Science and Technology

979-8-3195-3824-6/26/$31.00 ©2026 IEEE